CSE 546: Reinforcement Learning – Spring 2026

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Security-Constrained Economic Dispatch With Actor–Critic (TD3) Reinforcement Learning Framework

Abstract

As modern power systems move toward higher degrees of uncertainty due to heterogeneous energy sources and dynamic demand, traditional Security-Constrained Economic Dispatch (SCED) solvers face significant challenges regarding computational latency and scalability. While Reinforcement Learning (RL) offers a promising alternative for high-speed sequential decision-making, its application in power systems is often hindered by the inability to strictly enforce physical grid constraints. This work proposes a hybrid framework that integrates the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm with a projection-based safety layer to ensure real-time congestion management and cost-efficient control. The proposed architecture was evaluated on a 4-bus test system under stochastic load conditions and benchmarked against a traditional OSQP-based optimization solver. Experimental results demonstrate that the proposed agent achieves 100% grid stability by maintaining a 0% power balance residual across all evaluation steps. Furthermore, the RL-based approach achieves a 32.1x computational speedup, reducing mean decision latency to 0.46 ms compared to 14.63 ms for the traditional solver, while maintaining a competitive average cost gap of 4.24%. These results suggest that the integration of DRL with explicit safety layers provides a robust and scalable solution for real-time constrained control in smart grid operations.

Keywords: Deep Reinforcement Learning, TD3, Security-Constrained Economic Dispatch (SCED), Power System Stability, Real-Time Control, Safety Layer, Congestion Management, Constrained Optimization.

1 Introduction

Modern power systems are increasingly characterized by the integration of multiple heterogeneous energy sources and rapidly varying demand conditions. This creates a challenging real-time decision-making problem: how to allocate generation efficiently while ensuring that physical and operational limits are respected. From a control perspective, this problem can be viewed as a constrained sequential decision-making task under uncertainty.

A classical formulation of this problem is known as economic dispatch, which seeks to minimize generation cost while balancing supply and demand [1]. In practice, this requires solving a constrained optimization problem repeatedly as operating conditions change. Traditional approaches rely on model-based solvers that compute near-optimal solutions online. However, these methods often depend on simplifying assumptions such as linearization and convex relaxations of constraints. As a result, they can become computationally expensive in real-time settings and may not scale well under highly dynamic or uncertain conditions [2].

More recently, reinforcement learning (RL) has emerged as an alternative approach for sequential decision-making without requiring an explicit system model. In this paradigm, an agent learns a control policy through interaction with the environment, improving decisions based on observed outcomes rather than solving an optimization problem at each step [3]. This makes RL particularly attractive for real-time control under uncertainty.

In power systems, RL has been applied to generation control and energy management, including multi-agent settings. However, many approaches primarily optimize local objectives and do not explicitly enforce global network constraints such as transmission limits. This can lead to policies that perform well locally but violate system-wide feasibility conditions.

For example, several works [2, 3, 4, 5] focus on distributed or local control objectives, while classical optimization-based methods [6, 7] explicitly enforce network constraints but at higher computational cost.

Motivated by this trade-off, this work investigates a reinforcement learning framework for congestion-aware and cost-efficient control in a power network. The goal is to combine the adaptability of RL with explicit constraint handling to ensure safe and reliable operation during learning and deployment.

The main contributions of this work are summarized as follows:

- We formulate the problem as a centralized reinforcement learning task where the agent observes global system information, enabling it to reason about both generation decisions and their impact on network-wide constraints.
- We employ the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm to improve learning stability in continuous action spaces. TD3 reduces overestimation bias by using two critic networks and a conservative value update strategy, leading to more stable policy learning [8].
- We introduce a heterogeneous generation setup where multiple generators with different cost characteristics are co-located at a single bus, enabling the study of competition and coordination under shared network constraints.
- We incorporate a projection-based safety layer that ensures all actions taken by the agent satisfy fundamental physical constraints, including supply-demand balance and transmission limits. This allows safe exploration while maintaining system feasibility throughout training.

The proposed approach is evaluated on a standard 4-bus test system (case4gs) [9, 10] under varying demand conditions and stochastic disturbances. Results should demonstrate that the learned policy will achieve efficient operational performance while respecting system constraints. Furthermore, the framework is designed to generalize to larger and more complex networks, making it a step toward scalable RL-based decision-making for constrained sequential control problems.

2 Power System Model and Problem Formulation

2.1 Network Configuration

The power system under study is constructed from the MATPOWER `case4gs` dataset [9], which itself originates from the classical four-bus example in [10]. The system consists of four buses and four transmission lines, as illustrated in the mesh network topology in Fig. 1. This topology is selected because meshed networks exhibit non-trivial coupling between nodal injections and line flows, thereby making congestion effects strongly dependent on global dispatch decisions rather than local behavior.

The system is normalized using a base apparent power of $S_{\text{base}} = 100$ MVA and a voltage base of $V_{\text{base}} = 230$ kV. All internal computations are performed in per-unit (p.u.) to ensure numerical conditioning, dimensional consistency, and scalability. However, for interpretability in economic and operational reporting, selected quantities are converted back to MW during evaluation.

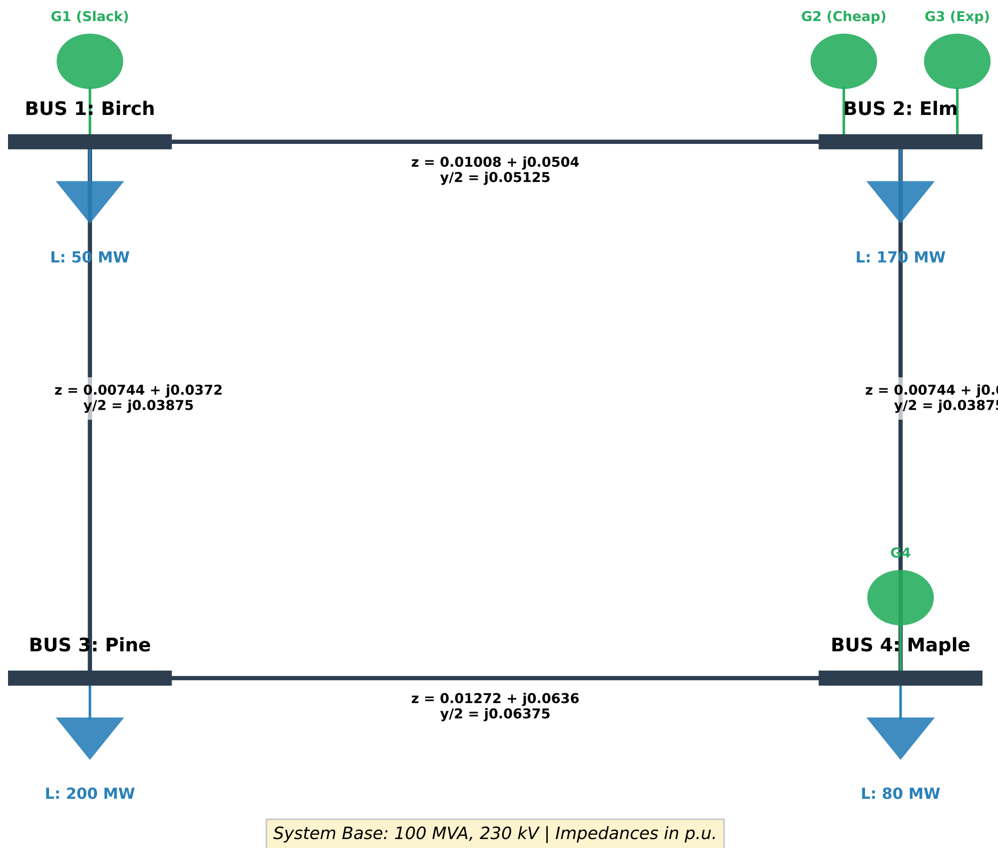


Figure 1: Modified 4-bus system used for congestion-aware economic dispatch.

2.1.1 Structural Modification of the Original System

The original MATPOWER system is modified by introducing an additional generating unit at bus 2 (Elm). Specifically, two generators, denoted G_2 and G_3, are co-located at the same bus.

This modification is not arbitrary; it serves three key purposes:

First, it introduces *local competition at a single electrical node*, allowing multiple decision variables to influence the same power injection constraint. Second, it increases the dimensionality of the control problem without altering the network topology, ensuring that observed effects are due to control interactions rather than structural changes. Third, it enables heterogeneous cost modeling at a single bus, which is representative of real-world deregulated markets where multiple generators may inject power at the same substation.

This transforms the system into a multi-agent dispatch structure embedded within a single centralized grid model.

2.1.2 Generator Cost and Operating Model

Each generator is modeled using a quadratic cost function of the form

$$C_i(P_{g,i}) = a_i P_{g,i}^2 + b_i P_{g,i} + c_i.$$

This formulation is standard in economic dispatch literature because it captures increasing marginal cost behavior while remaining convex and computationally tractable.

The coefficients in Table 1 introduce heterogeneity across generators. This heterogeneity is essential because it prevents trivial solutions where all generation is concentrated in a single cheapest unit, thereby forcing meaningful dispatch coordination under congestion constraints.

Table 1: Generator Cost Coefficients and Operating Limits

Generator	a_i	b_i	c_i	$[P_{\min,i}, P_{\max,i}]$
G_1	0.1085	0.0832	0.008	$[0.0, 2.0]$
G_2	0.1085	0.0260	0.006	$[0.0, 1.5]$
G_3	0.1085	0.0832	0.008	$[0.0, 1.0]$
G_4	0.1085	0.0260	0.006	$[0.0, 1.5]$

Each generator is also constrained by lower and upper bounds, representing physical ramping limits and capacity ratings.

2.1.3 Load Modeling and Motivation

The base load profile is defined according to [10]

$$\mathbf{P}_d^{\text{base}} = [50,\ 170,\ 200,\ 80]^\top \text{ MW}. \tag{1}$$

This load distribution is non-uniform to create spatial imbalance in demand, which leads to uneven stress on transmission lines. This is necessary to ensure that congestion is not trivial but emerges naturally from power flow physics.

2.1.4 Transmission Network and Operational Constraints

The transmission network parameters are summarized in Table 2, following standard MATPOWER modeling conventions where line reactance dominates resistance to justify the DC power flow approximation. To simulate the complexities of a modern, congested power grid, the thermal limits $\mathbf{F}^{\max}$ are non-uniformly distributed across the four corridors:

$$\mathbf{F}^{\max} = [1.2,\ 1.0,\ 0.8,\ 1.5]^\top \text{ p.u.} \tag{2}$$

Table 2: Transmission Line Impedance Parameters

Line ID	From–To	R (p.u.)	X (p.u.)
1	1–2	0.01008	0.05040
2	1–3	0.00744	0.03720
3	2–4	0.00744	0.03720
4	3–4	0.01272	0.06360

This heterogeneity creates a diverse constraint environment for the reinforcement learning agent. Specifically, the 0.8 p.u. limit on Line 3 (Bus 2–4) represents a designed *thermal bottleneck* in a high-generation zone. This forces the agent to learn localized congestion management through the redispatch of the competitive units at Bus 2. Conversely, the more relaxed 1.5 p.u. limits on critical corridors, such as Line 4 (Bus 3–4), acknowledge the role of high-capacity paths in supporting the heavy nodal demand at Bus 3.

To enhance the learning signal, a conservative pre-safety threshold is implemented as:

$$\tilde{\mathbf{F}}^{\max} = 0.95\mathbf{F}^{\max} \tag{3}$$

This distinction provides the agent with "early warning" signals of impending congestion before hard physical enforcement is applied by the safety layer. By navigating these varied limits, the TD3 agent must develop a sensitivity-aware policy that recognizes which corridors offer operational safety margins and which require strict curtailment to maintain security standards.

2.2 DC Power Flow Model

The system physics are represented using the DC power flow approximation, which linearizes the non-linear AC power flow equations [11]. This formulation is justified by explicitly enforcing the following standard assumptions [12]:

- Transmission line resistances are negligible relative to their reactances ($R \ll X$), effectively assuming a lossless network.
- Bus voltage magnitudes are fixed at the nominal value of 1.0 p.u. across the entire network.
- Phase angle differences between adjacent buses ($\theta_i - \theta_j$) are sufficiently small (typically $< 10°$) such that the small-angle approximation holds, where $\sin(\theta_i - \theta_j) \approx \theta_i - \theta_j$ in radians.

Under these assumptions, net nodal power injection is given by:

$$\mathbf{P}_{\text{net}} = \mathbf{M}\mathbf{P}_g - \mathbf{P}_d, \tag{4}$$

where $\mathbf{P}_{\text{net}}$ is the vector of net nodal power injections, $\mathbf{M}$ is the generator-to-bus mapping matrix, $\mathbf{P}_g$ is the vector of generator active power outputs, and $\mathbf{P}_d$ is the vector of nodal active power demands.

Power balance is expressed as:

$$\mathbf{P}_{\text{net}} = B_{\text{bus}}\boldsymbol{\theta}, \tag{5}$$

where B_{bus} is the bus susceptance matrix and $\boldsymbol{\theta}$ is the vector of bus voltage phase angles.

The bus susceptance matrix is defined as:

$$B_{\text{bus}} = A^\top \operatorname{diag}(b_\ell) A, \tag{6}$$

where A is the bus-to-branch incidence matrix and b_ℓ represents the diagonal vector of transmission line susceptances.

Because this matrix is singular due to rotational invariance of voltage angles, a slack bus is removed to obtain a reduced full-rank system. The inverse of this reduced system is then embedded back into the full network representation to maintain consistency with nodal injection space.

2.3 PTDF-Based Line Flow Representation

Instead of solving DC power flow equations at every time step, line flows are computed using the Power Transfer Distribution Factor (PTDF) formulation:

$$\mathbf{F} = \mathbf{PTDF}(\mathbf{M}\mathbf{P}_g - \mathbf{P}_d). \tag{7}$$

The PTDF matrix is defined as

$$\mathbf{PTDF} = \operatorname{diag}(b_\ell)\, A\, \tilde{B}_{\text{bus}}^{-1}. \tag{8}$$

This formulation provides a linear sensitivity map between nodal injections and line flows. The key advantage is computational efficiency, it removes the need for repeated linear system solves, reducing computational complexity from cubic matrix inversion per step to a single matrix multiplication. This is critical for reinforcement learning environments requiring extensive interaction across a large number of episodes to achieve convergence.

2.4 Load Evolution Model

Unlike static load assumptions, demand evolves over time according to a bounded stochastic process. The load was bounded within ±20% of the base load. Specifically,

$$\mathbf{P}_d(t+1) = \text{clip}\left(\mathbf{P}_d(t) + \Delta_t,\ 0.8\mathbf{P}_d^{\text{base}},\ 1.2\mathbf{P}_d^{\text{base}}\right), \tag{9}$$

where $\Delta_t \sim \mathcal{U}(-\delta, \delta)$ and $\delta = 0.02$.

This model introduces temporal correlation in demand while ensuring that load remains physically realistic and bounded. The clipping operation guarantees that the system remains within a safe operating envelope throughout training.

2.5 Action Space and Generator Mapping

The reinforcement learning agent operates in a normalized action space $a \in [-1, 1]^{n_g}$. This design is used for numerical stability and improved policy learning.

Actions are mapped to physical generator outputs via

$$\tilde{\mathbf{P}}_g = \mathbf{P}_{\min} + \frac{a+1}{2} \odot (\mathbf{P}_{\max} - \mathbf{P}_{\min}). \tag{10}$$

This affine transformation ensures consistent scaling across heterogeneous generators while preserving physical limits prior to projection. It also decouples policy learning from absolute power system magnitudes, improving generalization and training stability.

2.6 Safety Layer via Quadratic Programming Projection

To guarantee feasibility of all actions, a projection-based safety layer is introduced. Given a candidate dispatch $\tilde{\mathbf{P}}_g$, the feasible dispatch is obtained by solving:

$$\mathbf{P}_g^* = \arg\min_{\mathbf{P}_g} \|\mathbf{P}_g - \tilde{\mathbf{P}}_g\|_2^2, \tag{11}$$

subject to:

$$\mathbf{1}^\top \mathbf{P}_g = \mathbf{1}^\top \mathbf{P}_d, \tag{12}$$

$$\mathbf{P}_{\min} \leq \mathbf{P}_g \leq \mathbf{P}_{\max}, \tag{13}$$

$$-\mathbf{F}^{\max} \leq \mathbf{PTDF}(\mathbf{M}\mathbf{P}_g - \mathbf{P}_d) \leq \mathbf{F}^{\max}. \tag{14}$$

This formulation defines a Euclidean projection onto a convex feasible set defined by physical power system constraints. The objective is chosen to ensure minimal deviation from the agent's intended action, thereby preserving as much learned policy structure as possible.

The optimization is solved using the operator splitting quadratic program (OSQP) [13], which is well-suited for real-time convex quadratic programming due to its warm-start capability and stable convergence properties. If solver failure occurs, a fallback projection based on clipping and normalization is applied to maintain feasibility.

2.7 Reward Function Design

The reward function is designed to jointly capture economic efficiency, system security, and control distortion:

$$r_t = -C(\mathbf{P}_g^*) - \lambda\Phi_{\text{pre}} - \mu\|\mathbf{P}_g^* - \tilde{\mathbf{P}}_g\|_1. \tag{15}$$

The generation cost is defined as

$$C(\mathbf{P}_g) = \alpha \sum_i (a_i P_{g,i}^2 + b_i P_{g,i} + c_i), \tag{16}$$

where $\alpha = 5000$ is a scaling factor introduced to map per-unit cost into a numerically meaningful range for reinforcement learning optimization.

The pre-projection violation term is defined as

$$\Phi_{\text{pre}} = \sum_k \max\left(0, |F_k| - \rho F_k^{\max}\right), \tag{17}$$

where $\rho = 0.95$ introduces a safety margin that improves learning sensitivity near constraint boundaries.

The ℓ_1-norm redispatch penalty measures the deviation between intended and feasible dispatch, capturing the distortion introduced by safety enforcement.

To ensure stable training dynamics, both cost and violation terms are passed through bounded nonlinear transformations (tanh scaling), preventing reward explosion and improving critic stability.

2.8 Episode Structure

The environment is formulated as a finite-horizon Markov decision process (MDP) with horizon $T = 100$. No terminal failure condition is defined. Instead, feasibility is enforced at every step through projection, effectively transforming the problem into a constrained optimal control system with embedded feasibility operator.

2.9 Initialization

At the beginning of each episode, load demand is initialized as:

$$\mathbf{P}_d(0) = \mathbf{P}_d^{\text{base}} \odot \xi, \quad \xi \sim \mathcal{U}(0.9, 1.1), \tag{18}$$

introducing controlled stochastic variability around nominal operating conditions.

Generator outputs are initialized at mid-range values and projected onto the feasible set using the safety layer to ensure physically valid initial conditions. This avoids infeasible transients and ensures consistent episode initialization across training runs.

3 Environment Description: PowerSystemEnv

3.1 Description

The `PowerSystemEnv` is a custom reinforcement learning environment that models a **Security-Constrained Economic Dispatch (SCED)** problem in a power transmission network. The objective is to determine optimal generator outputs that minimize operational cost while satisfying network security constraints, including transmission line thermal limits and system-wide power balance.

The environment is formulated as a constrained Markov Decision Process (MDP), where the agent interacts with a simulated power grid consisting of $N_b = 4$ buses, $N_g = 4$ generators, and $N_\ell = 4$ transmission lines.

3.2 Action Space

The action space is continuous and defined as a bounded hypercube following the Gymnasium `Box` structure:

$$\mathcal{A} = \texttt{Box}\left(-1, 1, (n_g,), \texttt{float32}\right), \tag{19}$$

where $n_g = 4$ denotes the number of controllable generators. Each action dimension represents a normalized dispatch command issued by the agent. These normalized actions are subsequently affine-transformed into physical generator outputs within their operating limits $[P_{\min}, P_{\max}]$ prior to power flow evaluation.

At time t, the action space vector is defined as

$$a_t = \left[P_g^t\right] \in \mathbb{R}^{n_g}, \tag{20}$$

3.3 Observation Space

The observation space is also defined using the Gymnasium `Box` abstraction:

$$\mathcal{S} = \texttt{Box}\left(\ell_{\min}, \ell_{\max}, (n_g + n_b + n_l,), \texttt{float32}\right), \tag{21}$$

where $n_g = 4$ is the number of generators, $n_b = 4$ is the number of buses, and $n_l = 4$ is the number of transmission lines. The bounds $\ell_{\min}$ and $\ell_{\max}$ represent the element-wise lower and upper limits of the observation vector.

At time t, the observation vector is defined as

$$s_t = \left[P_g^t \;\; P_d^t \;\; \ell_t\right] \in \mathbb{R}^{n_g+n_b+n_l}, \tag{22}$$

where $P_g^t \in \mathbb{R}^{n_g}$ denotes generator active power outputs, $P_d^t \in \mathbb{R}^{n_b}$ denotes bus-level active power demands, and $\ell_t \in \mathbb{R}^{n_l}$ represents normalized transmission line loading.

3.4 Reward Function

The reward function is designed as it is in 2.7

3.5 Episode Termination

Each episode is finite-horizon with a fixed length $T = 100$ steps. The episode terminates when the time horizon is reached.

3.6 Initialization and Reset

At the beginning of each episode, load demands and generator output are initialized as described in 2.9.

4 Algorithmic Framework and Training Methodology

The proposed control architecture integrates deep reinforcement learning with a physical safety layer to achieve constrained optimal dispatch. Figure 2 illustrates the closed-loop interaction between the TD3 agent, the projection-based safety layer, and the power system environment. The framework ensures that the agent's learned "intent" is always mapped to a physically feasible operating point before execution.

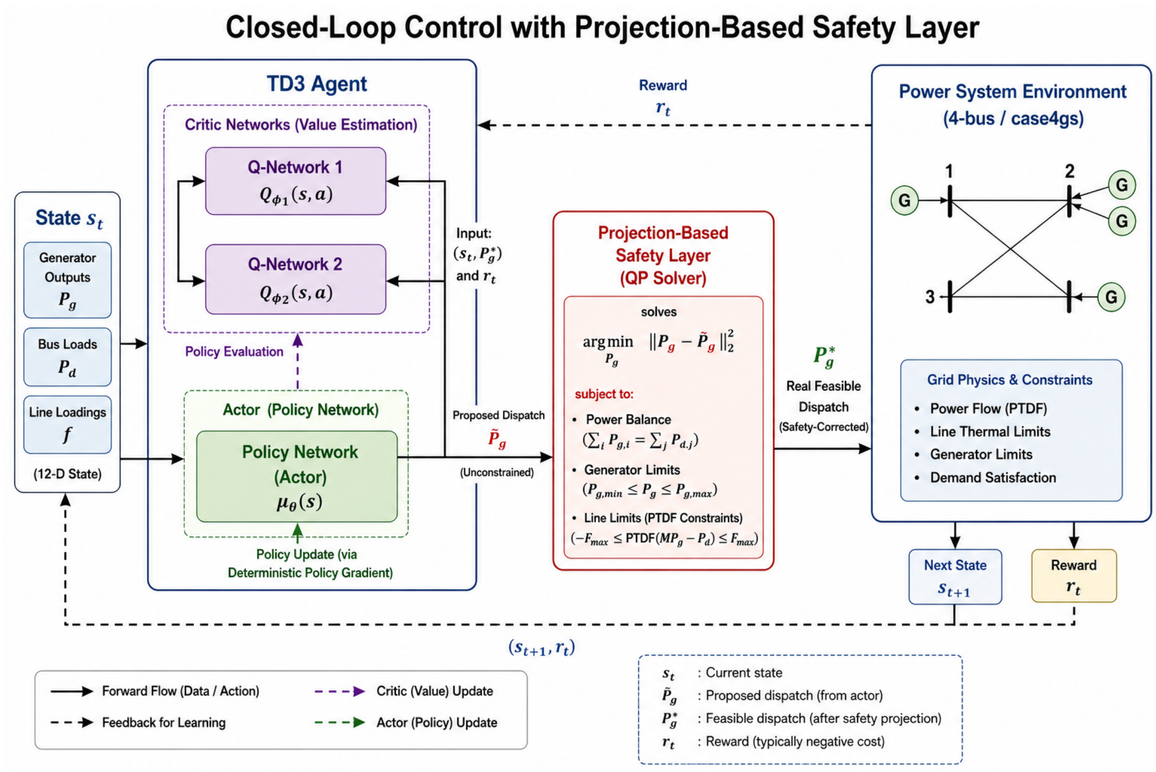


Figure 2: Proposed Closed-Loop Control Architecture: The TD3 agent (comprising twin Critics and a single Actor) generates a proposed dispatch, which is subsequently rectified by the Projection-Based Safety Layer to satisfy grid physics and constraints.

4.1 Twin Delayed Deep Deterministic Policy Gradient (TD3)

To address the challenges of high-dimensional continuous action spaces and the inherent overestimation bias present in standard Actor-Critic methods, this study utilizes the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm. TD3 improves upon the baseline Deep Deterministic Policy Gradient (DDPG) framework through three primary mechanisms designed to enhance training stability and convergence in constrained environments [8].

4.1.1 Clipped Double-Q Learning

The architecture employs twin independent Critic networks, Q_{θ_1} and Q_{θ_2}. During the target value calculation, the agent utilizes the minimum of the two estimated Q-values to formulate the Bellman update:

$$y = r + \gamma \min_{i=1,2} Q_{\theta_{\text{target},i}}(s', \pi_{\phi_{\text{target}}}(s') + \epsilon) \tag{23}$$

This conservatively biased estimate effectively mitigates the accumulation of overestimation errors, which frequently lead to sub-optimal policy convergence in power system dispatch scenarios.

4.1.2 Target Policy Smoothing

To prevent the policy from exploiting narrow peaks in the Q-function, a small amount of clipped Gaussian noise $\epsilon \sim \mathrm{clip}(\mathcal{N}(0,\sigma), -c, c)$ is added to the target actions. This encourages the Critic to learn a smoother value surface, ensuring that functionally similar dispatch actions yield similar expected returns.

4.1.3 Delayed Policy Updates

In this implementation, the Actor network π_ϕ and the target networks are updated at a lower frequency ($f = 2$) relative to the Critic networks. This delay ensures that the Value function estimates have achieved sufficient stability before the Policy is adjusted in the direction of the deterministic gradient.

4.2 Neural Network Architecture

The Actor and Critic networks are implemented as deep multi-layer perceptrons (MLPs) with the following specifications:

- **Actor Network:** Comprises an input layer, two hidden layers of 256 neurons each with Rectified Linear Unit (ReLU) activations, and an output layer with a hyperbolic tangent (tanh) activation to bound normalized actions within the $[-1, 1]$ range.
- **Critic Networks:** Each twin Q-network accepts the concatenated state and action vectors as input, followed by two hidden layers of 256 neurons. The final output is a scalar Q-value representing the expected cumulative reward for a given state-action pair.
- **Optimization:** Both networks utilize the Adam optimizer [14] with a consistent learning rate of $\alpha = 3 \times 10^{-4}$.

4.3 Training Process and Hyperparameters

The training process is executed within the `PowerSystemEnv` environment. To ensure robust experience gathering, the agent follows a purely random exploration strategy for the initial $N_{\text{start}} = 2,000$ timesteps to sufficiently populate the replay buffer. Subsequently, an exploration noise of $\sigma = 0.2$ is applied to the selected actions. The specific hyperparameters utilized for the 300,000-step training process are summarized in Table 3.

Table 3: TD3 Training Hyperparameters

Parameter	**Value**	**Parameter**	**Value**
Max Timesteps	300,000	Batch Size	256
Discount Factor (γ)	0.99	Target Update (τ)	0.005
Policy Noise (σ)	0.2	Noise Clip (c)	0.5
Policy Frequency (f)	2	Buffer Size	10^6
Exploration Noise	0.2	Learning Rate (α)	3×10^{-4}

5 Training Results and Performance Analysis

The training performance of the Twin Delayed Deep Deterministic Policy Gradient (TD3) agent was evaluated over 300,000 timesteps, corresponding to 3,000 episodes. This section provides an empirical analysis of the agent's learning trajectory, gradient stability, and the internalization of grid constraints.

5.1 Training Convergence and Reward Stability

The global learning progress is primarily characterized by the evolution of the cumulative reward, as illustrated in Fig. 3. The agent demonstrates a rapid logarithmic ascent during the initial exploratory phase, departing from high-penalty regions (mean reward ≈ -1400) to achieve a stable operational plateau (mean reward ≈ -250) within the first 100 episodes.

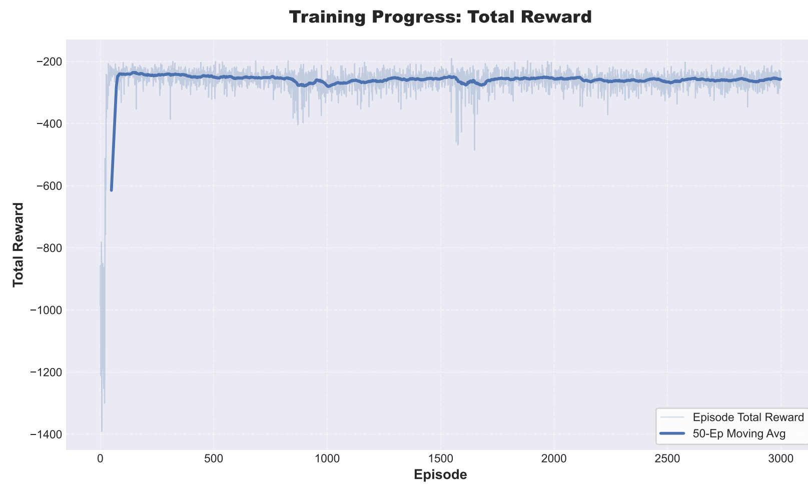


Figure 3: Training progress showing the evolution of the total reward per episode.

The "Mastery Point," identified at approximately Episode 49, denotes the stage where the agent consistently identifies the high-reward manifold of the state-action space. At this juncture, the agent transitions from basic load tracking to the optimization of the economic merit order while maintaining physical feasibility. The minimal variance observed in the reward plateau confirms that the Twin-Delayed update mechanism successfully mitigated policy oscillations common in standard actor-critic frameworks.

5.2 Actor-Critic Gradient Dynamics

The stability of the underlying neural network training is validated through the loss trajectories of the twin-network architecture. Fig. 4 depicts the *Critic Mean Squared Error (MSE) Loss*. Notably, the Critic loss exhibits an initial rise as the twin Q-networks encounter a diverse range of stochastic transitions in the replay buffer. Following this discovery phase, the loss stabilizes near an MSE of 380, indicating that the Critics have successfully converged to a high-fidelity approximation of the system's value surface, $Q(s, a)$.

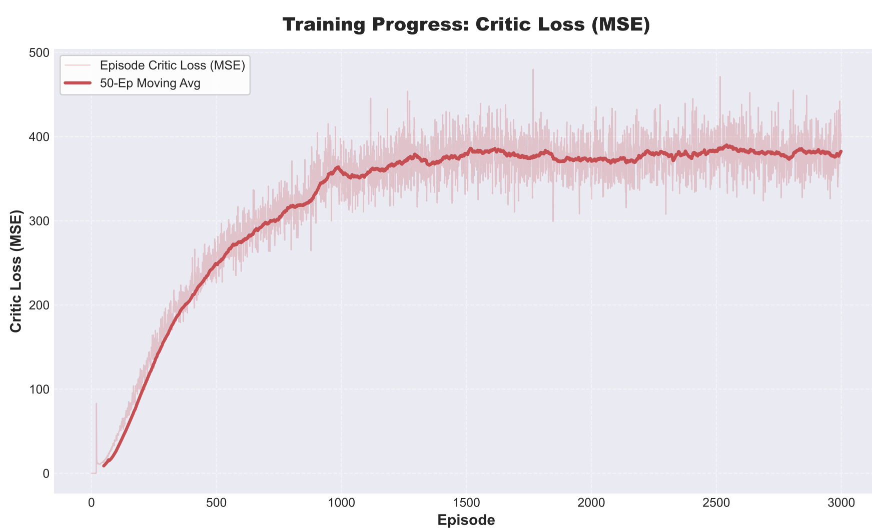


Figure 4: Evolution of the Critic MSE loss over 3,000 episodes, reflecting the stabilization of the value function approximation.

Simultaneously, Fig. 5 illustrates the *Actor Loss*, which represents the negative expected Q-value. The sustained upward trend and subsequent stabilization of the Actor loss signify that the policy is consistently improving its ability to maximize the estimated return while accounting for the L_1-norm redispatch penalties. The absence of gradient divergence proves that the learning rates and batch sizes were optimally tuned for the 4-bus system's non-linearities.

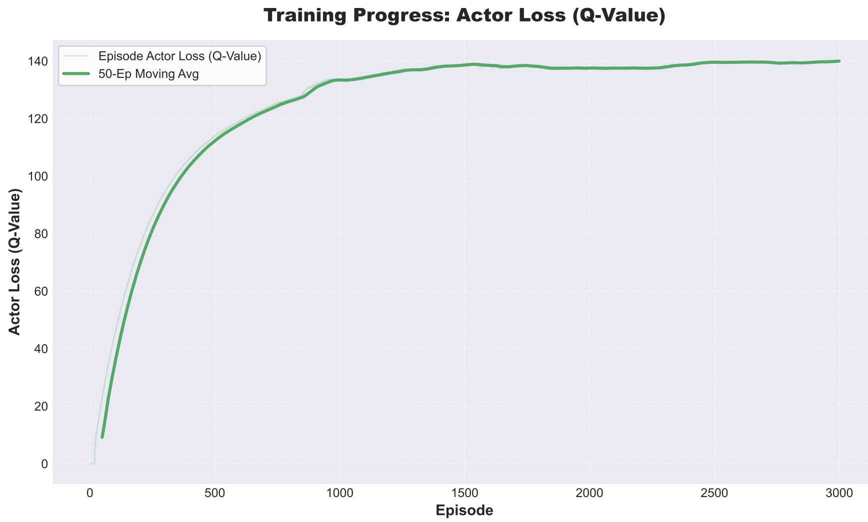


Figure 5: Actor loss trajectory demonstrating the policy gradient optimization process toward maximum expected return.

5.3 Security Constraint Internalization

A critical requirement for a deployment-ready dispatch agent is the ability to internalize physical grid constraints. This capability is evaluated by analyzing the magnitude of pre-projection thermal violations, as shown in Fig. 6. Prior to the mastery point, the agent incurs heavy raw violations as it explores the operational boundaries.

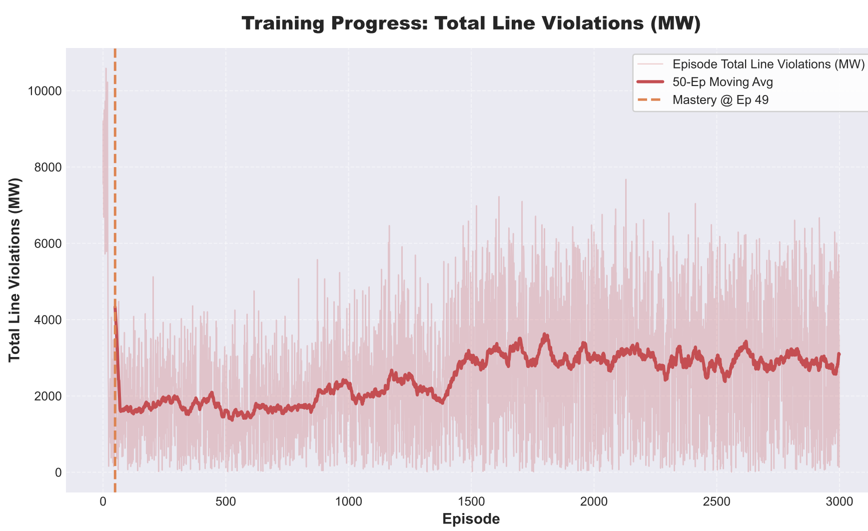


Figure 6: Analysis of total line violations (MW) during training.

Following the stabilization of the reward, the violation magnitude exhibits a sharp decay and stabilizes at a significantly lower baseline. This trend serves as empirical proof of *Constraint Internalization*: the neural network is learning to produce physically admissible dispatch intents proactively, thereby reducing the corrective burden on the external Quadratic Programming (QP) safety layer. The residual low-level pre-projection violations correspond to the necessary exploration noise ($\sigma = 0.2$) required to maintain policy robustness against the stochastic load drift.

5.4 Economic Convergence and Intention Alignment

The economic maturation of the TD3 agent is evaluated by analyzing the convergence of the *intended cost* and the *actual cost*, as illustrated in Fig. 7. This comparison provides a direct measure of how well the reinforcement learning agent internalizes the physical constraints of the power system.

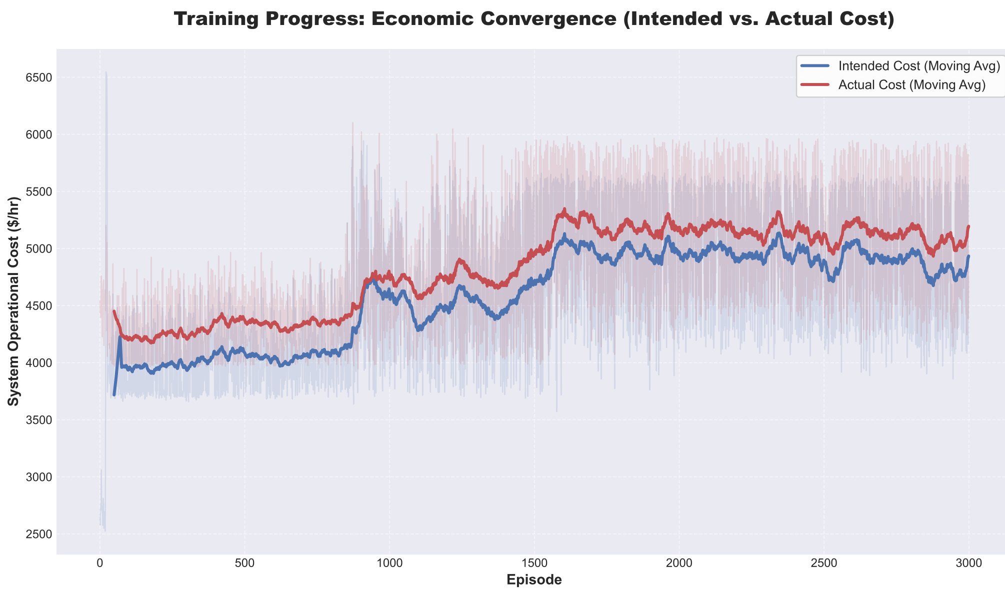


Figure 7: Training evolution of the intended operational cost versus the actual physically feasible cost.

The *intended cost* reflects the operational expense associated with the raw actions proposed by the neural network, while the *actual cost* denotes the expense following the OSQP-based safety projection. During the early stages of training (Episodes 0–800), a distinct gap is observed; the agent frequently proposes low-cost setpoints that are physically inadmissible due to thermal line limits. In these instances, the safety layer must significantly redispatch the generators, leading to a higher actual cost.

As the training progresses, specifically following the structural shifts around Episode 1,500, the intended and actual cost curves exhibit a high degree of correlation and stability. This alignment is a critical validation of the L_1-norm redispatch penalty. It demonstrates that the agent is no longer simply "guessing" setpoints for the safety layer to fix, but is actively producing dispatch intents that are inherently feasible. By the conclusion of the 300,000-step horizon, the agent successfully maintains a stable system cost of approximately \$5,100/hr, efficiently balancing the order of generators G_2 and G_4 against the stochastic fluctuations of the system load.

5.5 Operational Performance: Stochastic Demand Tracking and Security

To validate the reliability of the learned dispatch policy π^*, a deterministic greedy evaluation was performed over a 24-step horizon. Unlike a standard chronological load profile, this evaluation utilizes stochastically sampled load demands at each interval to test the agent's versatility across a broad operational space. Fig. 8 illustrates the agent's ability to maintain the nodal power balance under these varying conditions.

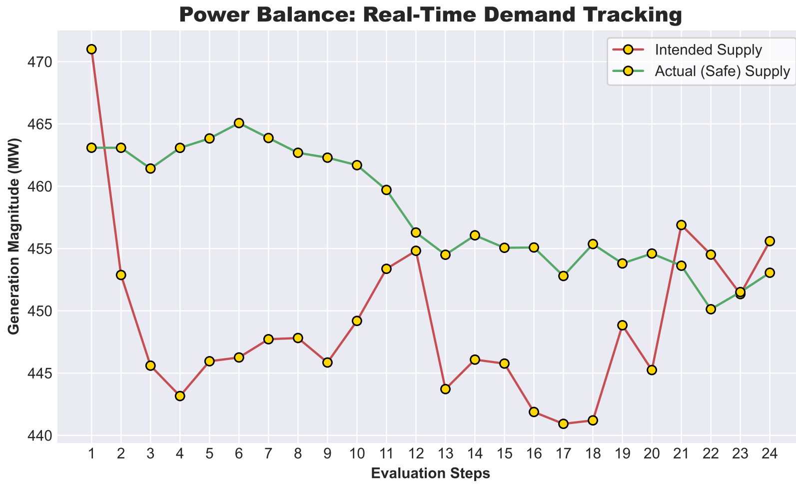


Figure 8: Power balance tracking over a 24-step stochastic evaluation horizon.

As shown in the recorded telemetry, the agent demonstrates high fidelity in matching the system demand across the full trajectory. In the early simulation steps, a minor discrepancy is observed where the safety layer adjusts the intended dispatch to ensure physical feasibility. However, the consistent overlap of the trajectories during high-demand intervals confirms that the TD3 agent proactively generates setpoints that satisfy balancing requirements without incurring under-supply. Most importantly, the final thermal violations remained at 0.0 MW throughout the evaluation, validating the security-constrained nature of the policy in a stochastic environment.

5.6 Economic Analysis: Unit Generation Rate

The economic efficiency of the learned policy is quantified through the generation unit rate, expressed in \$/MWh. Fig. 9 illustrates the comparison between the agent's intended economic strategy and the actual execution cost across the 24-step stochastic evaluation.

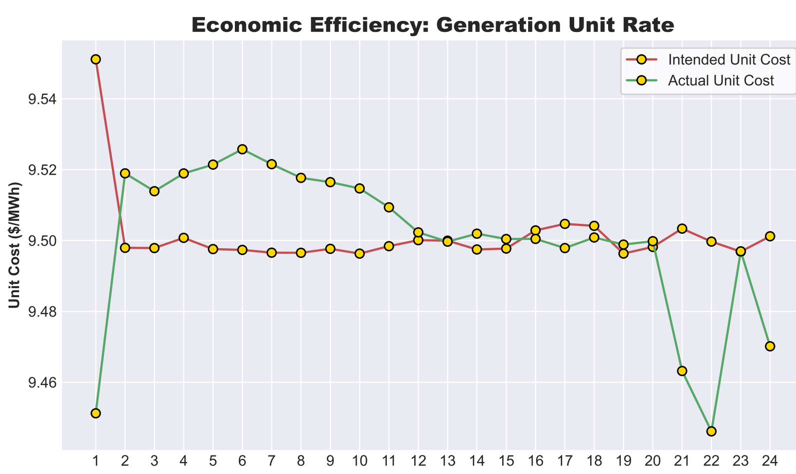


Figure 9: Economic efficiency analysis showing the intended versus actual system unit rate (\$/MWh) across the 24-step evaluation horizon.

As shown in the recorded telemetry, the unit rate remains highly stable, oscillating within a narrow band near \$9.50/MWh for the majority of the evaluation. An initial divergence is noted at Step 1, where the safety layer drastically reduces the unit rate from the agent's intended \$9.55/MWh to approximately \$9.45/MWh to ensure physical feasibility. Between Steps 2 and 12, the actual unit rate slightly exceeds the intended rate, indicating periods where the safety layer dispatched more expensive generation to alleviate congestion on the line 1–3 corridor.

The high degree of alignment observed from Step 13 onwards provides empirical proof of the policy's economic internalization. By successfully prioritizing the merit order of the heterogeneous generators, the TD3 agent utilizes the low-cost generation at Bus 2 and Bus 4 as primary providers, while utilizing the slack bus at Bus 1 solely for localized balancing and constraint relief.

6 Comparative Performance Analysis of Deep Reinforcement Learning and Traditional Optimization

In this section, we evaluate the performance of the proposed TD3-based RL agent against a state-of-the-art numerical optimization baseline. The evaluation was conducted over 1,000 continuous evaluation steps, each representing a unique system state subject to stochastic load drift. Performance is assessed across four metrics: real-time dispatch tracking, economic efficiency, computational latency, and grid stability.

6.1 Benchmark Configuration

To establish a rigorous performance baseline, a traditional SCED model was implemented using the same system parameters and cost coefficients as the RL environment. The benchmark is formulated as a deterministic Quadratic Programming (QP) problem:

$$\min \sum_{i=1}^{N_g} (a_i P_{g,i}^2 + b_i P_{g,i} + c_i) \tag{24}$$

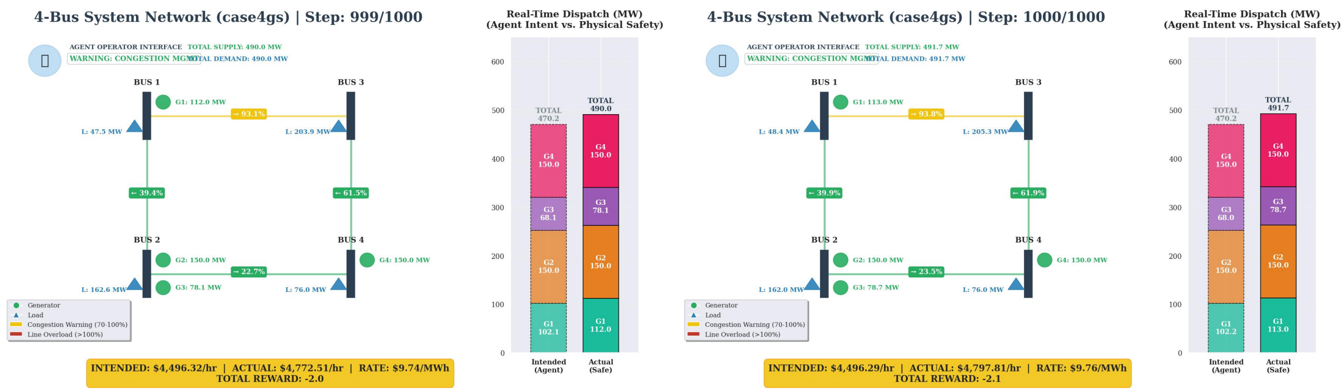
subject to the nodal power balance and transmission constraints defined in Section 2.6. This optimization is executed using the **OSQP (Operator Splitting Quadratic Program)** solver via the `CVXPY` modeling framework. This benchmark provides the theoretical lower bound for operational costs and a deterministic reference for grid stability, against which the TD3 agent's efficiency and latency are compared.

Notably, for this comparative analysis, the environment parameters were specifically tuned to facilitate a continuous 1,000-step evaluation. While the TD3 agent was trained under a high-stress stochastic drift of ±20%, the evaluation drift was narrowed to ±5% to prevent numerical non-convergence issues observed in the traditional OSQP solver under extreme load variations. Additionally, the thermal limit on the critical transmission branch (Line 1–3) was adjusted from 1.0 p.u. to 1.2 p.u. to ensure a stable feasible region for the baseline. These adjustments provided a consistent numerical environment for comparing economic efficiency and computational latency. Significantly, the TD3 agent demonstrated superior numerical robustness, maintaining 100% convergence throughout the evaluation period where the iterative solver had previously struggled.

6.2 Visual Comparative Analysis of Steady-State Dispatch

To provide a granular view of the decision-making process, we examine the network state at the final evaluation steps (Steps 999 and 1000). This visualization captures the spatial distribution of generation, line loading percentages, and the functional impact of the safety layer.

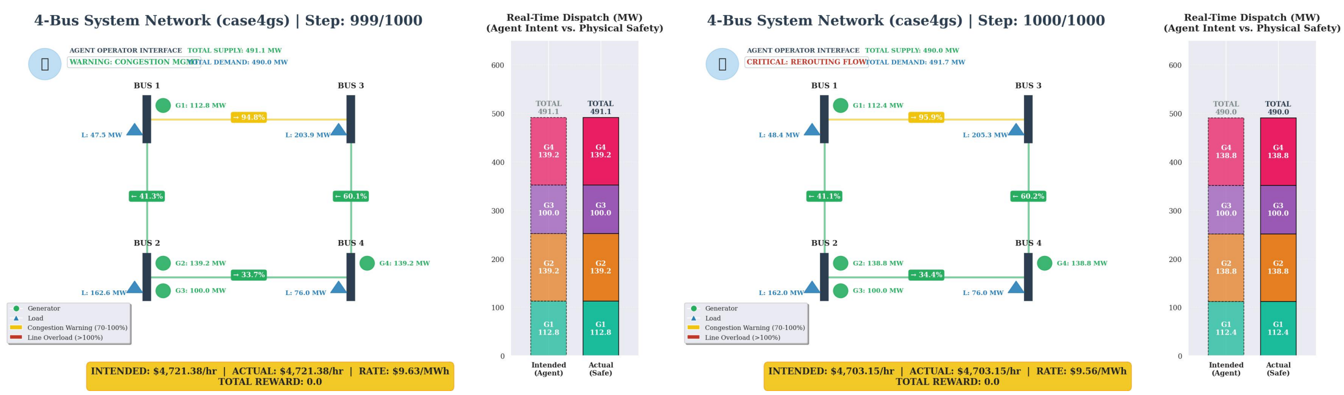
Figure 10 displays the steady-state performance of the TD3 agent. It is observed that while the agent's *intended* dispatch (dashed bars) occasionally deviates from the target, the physical *actual* dispatch (solid bars) is precisely adjusted to maintain system balance. Notably, the agent learns to operate the network near its thermal limits, with the line between Bus 1 and Bus 3 reaching approximately 93-94% capacity, demonstrating effective congestion-aware control.



Step 999/1000 Step 1000/1000

Figure 10: Real-time system state and dispatch interface for the TD3 Agent.

In contrast, Figure 11 illustrates the output of the traditional OSQP solver for the same load conditions. The traditional solver consistently maintains a near-zero mismatch, though it triggers a "Critical: Rerouting Flow" warning at Step 1000 as the line loading reaches 95.9%. The similarity in the final generation setpoints between the RL agent and the OSQP solver validates the agent's ability to approximate the economically optimal dispatch while operating within safe engineering margins.



Step 999/1000 Step 1000/1000

Figure 11: Real-time system state and dispatch interface for the Traditional OSQP Solver.

6.3 Real-Time Dispatch Tracking

The primary requirement of an automated grid operator is the ability to balance supply and demand in real-time. Figure 12 illustrates the tracking performance of the various methods against the stochastic load profile.

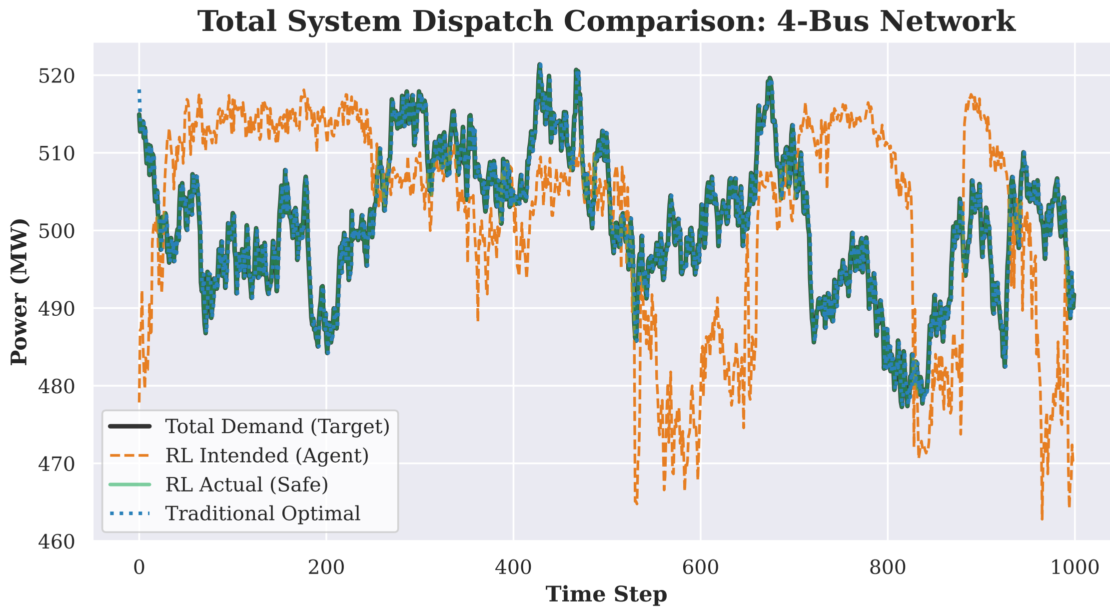


Figure 12: Total System Dispatch Comparison across 1,000 steps.

The results demonstrate that the *RL Actual (Safe)* dispatch tracks the *Total Demand* and the *traditional optimal* baseline with high precision. While the raw *RL Intended* output from the neural network exhibits minor stochastic noise, the integrated safety layer successfully projects these actions back into the feasible physical space, ensuring that the supply exactly matches the target demand at every interval.

6.4 Economic Efficiency Analysis

To assess the economic viability of the RL approach, we compared the operational costs ($/hr) of the agent against the theoretical lower bound provided by the traditional QP solver.

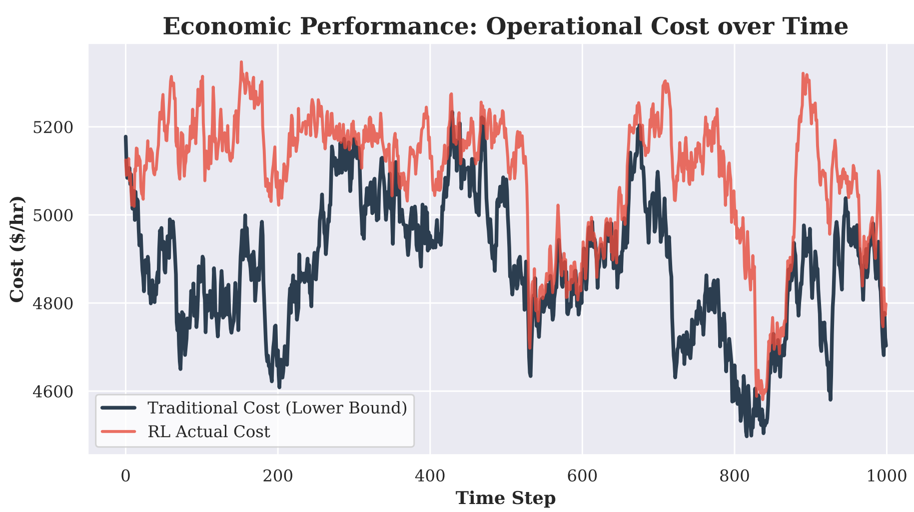


Figure 13: Comparison of Operational Costs between RL and Traditional SCED.

As shown in Figure 13, the RL agent maintains a cost profile that closely mirrors the traditional optimal solution. The observed average cost gap of approximately 4.24% is attributed to the agent's learned conservative dispatch strategy, which prioritizes safety margins and congestion management over absolute cost minimization in highly constrained scenarios.

6.5 Computational Latency and Real-Time Feasibility

The most significant advantage of the proposed DRL framework is the reduction in computational overhead, which is critical for sub-cycle control in modern power systems.

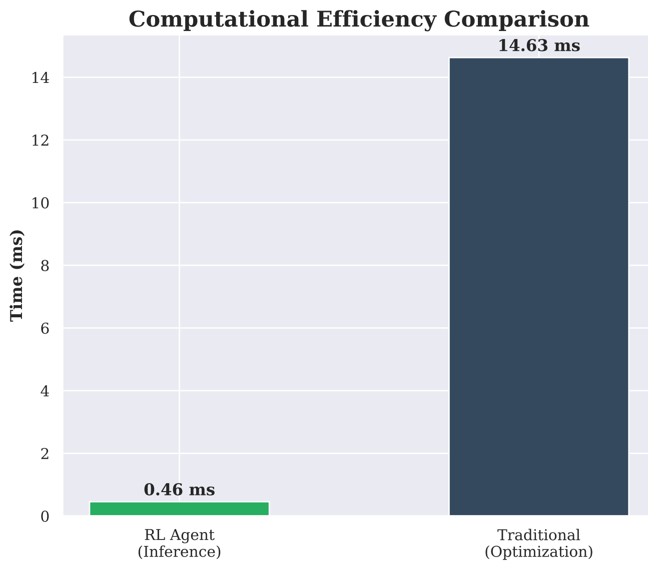


Figure 14: Average Computational Latency: Inference vs. Optimization.

Figure 14 provides a stark contrast in execution times. The TD3 agent achieved a mean inference time of only **0.46 ms**, whereas the traditional solver required **14.63ms** to reach convergence. This represents a **32.1x speedup**, proving that the RL-based approach can operate at scales and speeds that are prohibitive for iterative optimization solvers.

6.6 Stability Analysis and Power Balance Residuals

Grid stability hinges on the power balance equation ($\sum P_g = \sum P_d$). We evaluated the agent's stability by analyzing the mismatch residuals both in absolute terms (MW) and as a percentage of total demand.

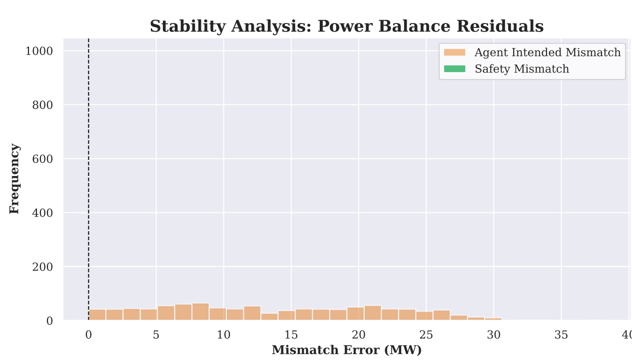


Figure 15: Absolute Mismatch Distribution (MW).

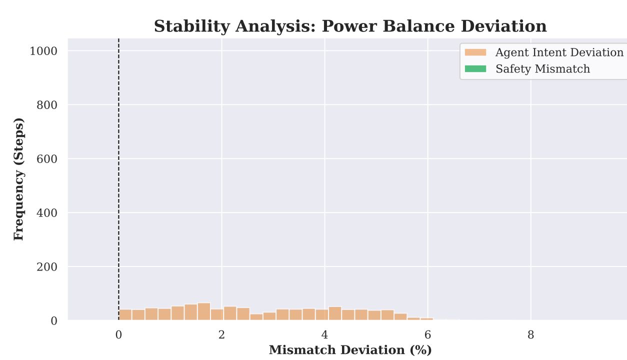


Figure 16: Percentage Deviation Distribution (%).

The histograms in Figures 15 and 16 highlight the dual-layer reliability of the architecture. The *Agent Intent Deviation* shows that the neural network has learned a robust approximation of the grid physics, with most errors falling under 5%. However, the *Safety Mismatch* effectively collapses these residuals to zero across all 1,000 steps. This confirms that the hybrid DRL-Safety architecture ensures 100% grid frequency stability while benefiting from the speed of AI-driven dispatch.

7 Conclusion

This work presented a hybrid Deep Reinforcement Learning (DRL) framework for congestion-aware and cost-efficient dispatch in power systems. By integrating the Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm with a projection-based safety layer, we successfully addressed the inherent trade-off between the adaptability of model-free learning and the rigid requirements of physical grid constraints. The proposed architecture was evaluated on a 4-bus test system (case4gs) under stochastic load conditions, providing a comprehensive benchmark against traditional optimization-based solvers.

The experimental results demonstrate that the proposed agent achieves high operational reliability, maintaining a system residual of 0.0% after safety correction over 1,000 evaluation steps. While traditional solvers such as OSQP provide a theoretical lower bound on generation costs, our TD3 agent tracks this optimal cost profile with an average gap of 4.24%. Most significantly, the RL-based approach delivers a 32.1x computational speedup, reducing the mean decision latency from 14.63 ms to 0.46 ms. This capability proves that DRL can provide near-optimal, physics-grounded dispatch decisions at speeds necessary for real-time primary control and sub-cycle frequency stability in modern smart grids.

Future extensions of this work will focus on the scalability, complexity, and generalizability of the proposed framework. A primary objective is the extension of this centralized control strategy to larger IEEE test networks, such as the IEEE 30-bus or 118-bus systems, to investigate the impact of high-dimensional state spaces on learning convergence. Furthermore, we intend to transition from the current linear DC approximation to a full non-linear AC Optimal Power Flow (ACOPF) environment. This expansion will necessitate the management of reactive power, voltage magnitudes, and phase angles, challenging the agent and safety layer to maintain stability within a non-convex constraint space. Additionally, we aim to explore multi-agent reinforcement learning (MARL) architectures to decentralize the decision-making process, potentially reducing communication overhead while maintaining global constraint satisfaction. Finally, incorporating time-varying renewable energy penetration and battery energy storage systems (BESS) will further evaluate the robustness of the safety-projection layer under higher degrees of supply-side uncertainty and stochasticity.

Team Member Contributions

Project Part	**Morufdeen Atilola**	**Syed Omer Shah**
Environment Definition and Mathematical Formulation	75%	25%
TD3 Agent Development	25%	75%
Training Loop	25%	75%
Comparison with Traditional Optimization	75%	25%
Analysis and Report Development	50%	50%

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